

w1 PID - Hybrid Swing Up model

PID tuning works nicely in a threshold error only $\rightarrow$ Below a certain point, movement becomes rather erratic or atleast $\rightarrow$ There's better ways to solve it

Swing-up $\rightarrow$ This model basically adds energy to the system

$$\tau = \dot{\theta} \cdot k_{\text{swing}} \quad \text{--- ① as } \dot{\theta} \Rightarrow \tau$$

Added condⁿ: when $\dot{\theta} < \text{Threshold}$ (basically say zero)
& is at bottom

Then τ via ① $= 0 \Rightarrow$ no movement

Hence, we add ext τ hardcoded for movement to start.